

SIMATS ENGINEERING

ASSESSMENT TOOL 3

NAME: O.Siva Charan
COURSE NAME: CSA1607

REG.NO: 192525039
COURSE CODE: Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO4: Examine the applicability of clustering algorithms in real-world scenarios, presenting findings through clear reports with effective visualizations.
Assessment Tool Weightage: 20%

QUESTIONS: 05

Total Marks:50

Annexure A

COMPARITIVE ANALYSIS

<i>Criteria</i>	Scope of Items (2)	Comparison Depth(2.5)	Use of Evidence (2)	Critical Thinking (2)	Clarity of Presentation (1)	<i>Total(10)</i>
<i>Weightage</i>	20%	25%	20%	20%	10%	<i>100%</i>
<i>Q1</i>						
<i>Q2</i>						
<i>Q3</i>						
<i>Q4</i>						
<i>Q5</i>						

Code of Conduct:

I _O.Siva Charan(192525039) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the
student:siva
charan

RUBRICS FOR CALCULATION:

		Comparative analysis			
Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Annexure A

Q1 . Dataset: Assessment Tool 3: Comparative Analysis (25% → 5 Marks)

Time duration :45 Minutes
Assessment Tool 3 (Low)

Q1 . Dataset: Customer Data

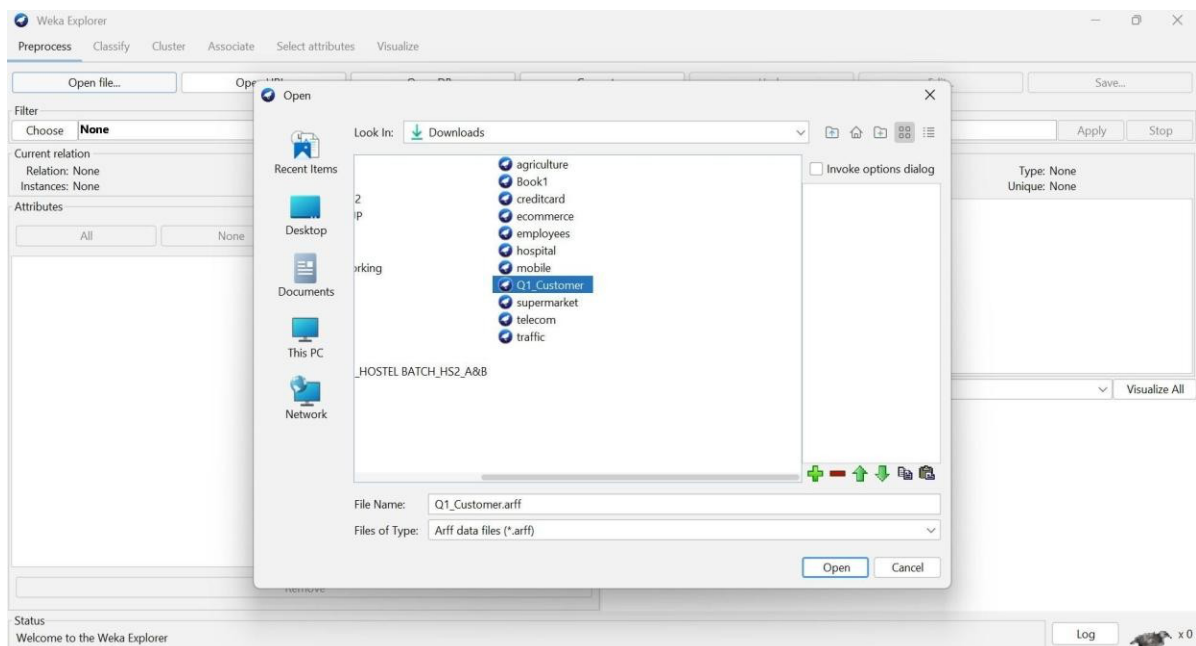
Customer Income (₹) Spending Score

C1	30000	60
C2	60000	40
C3	50000	70
C4	28000	65
C5	65000	35

Question:

Compare K-means and hierarchical clustering using this dataset and explain differences in performance and scalability. (BL3 – 1 Mark)

OUTPUT:



Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Cluster

Choose **HierarchicalClusterer - N 2 - L SINGLE - P - A "weka.core.EuclideanDistance - R first-last"**

Cluster mode

☒ Use training set

☐ Supplied test set

☐ Percentage split

☐ Classes to clusters evaluation

(Num) SpendingScore

☒ Store clusters for visualization

Ignore attributes

Start Stop

Result list (right-click for options)

06:38:09 - SimpleKMeans

06:38:50 - HierarchicalClusterer

Clusterer output

Initial starting points (random):

Cluster 0: C4,28000,65

Cluster 1: C2,60000,40

Missing values globally replaced with mean/mode

Final cluster centroids:

Attribute	Full Data	Cluster#	0	1
	(5.0)	(3.0)	(2.0)	
Customer	C1	C1	C2	
Income	46600	36000	62500	
SpendingScore	54	65	37.5	

Time taken to build model (full training data) : 0.02 seconds

=== Model and evaluation on training set ===

Clustered Instances

0	3 (60%)
1	2 (40%)

Status OK

Log

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Cluster

Choose **HierarchicalClusterer - N 2 - L SINGLE - P - A "weka.core.EuclideanDistance - R first-last"**

Cluster mode

☒ Use training set

☐ Supplied test set

☐ Percentage split

☐ Classes to clusters evaluation

(Num) SpendingScore

☒ Store clusters for visualization

Ignore attributes

Start Stop

Result list (right-click for options)

06:38:09 - SimpleKMeans

06:38:50 - HierarchicalClusterer

Weka Clusterer Visualize: 06:38:09 - SimpleKMeans (customer_clustering)

X: Instance_number (Num)

Y: Customer (Nom)

Colour: Cluster (Nom)

Reset Clear Open Save

Jitter

Plot: customer_clustering_clustered

Class colour

cluster0 cluster1

Status OK

Log

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Open file...

Open

Look In: Downloads

Recent Items

2

p

Desktop

Working

Documents

This PC

Network

Invoke options dialog

Type: None

Unique: None

Visualize All

File Name: Q1_Customer.arff

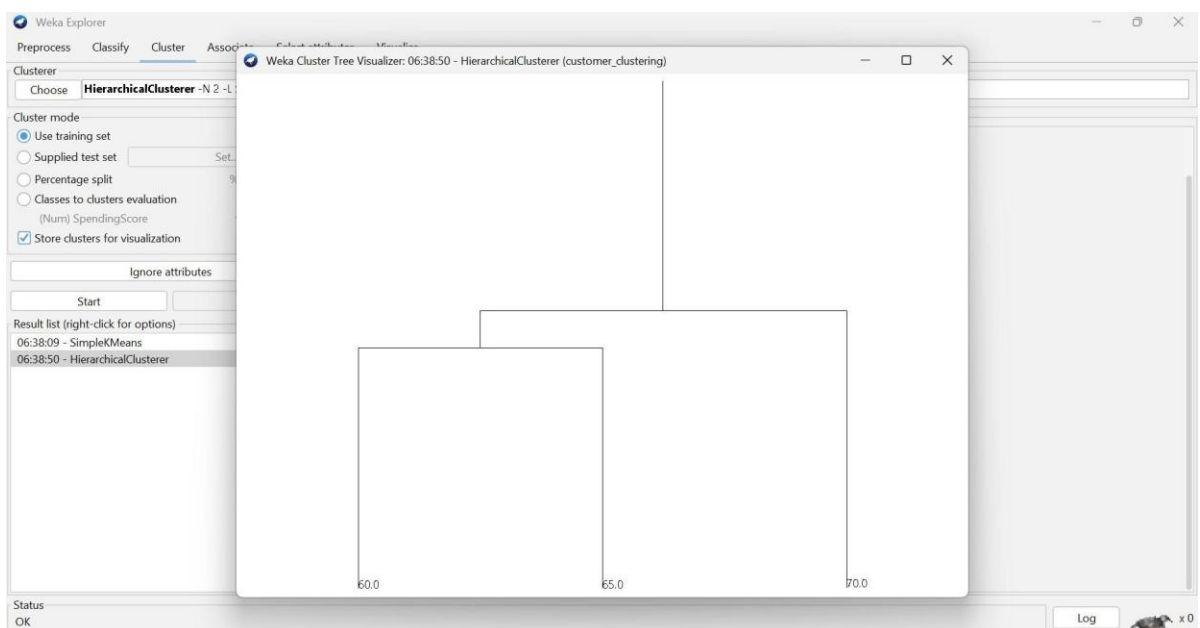
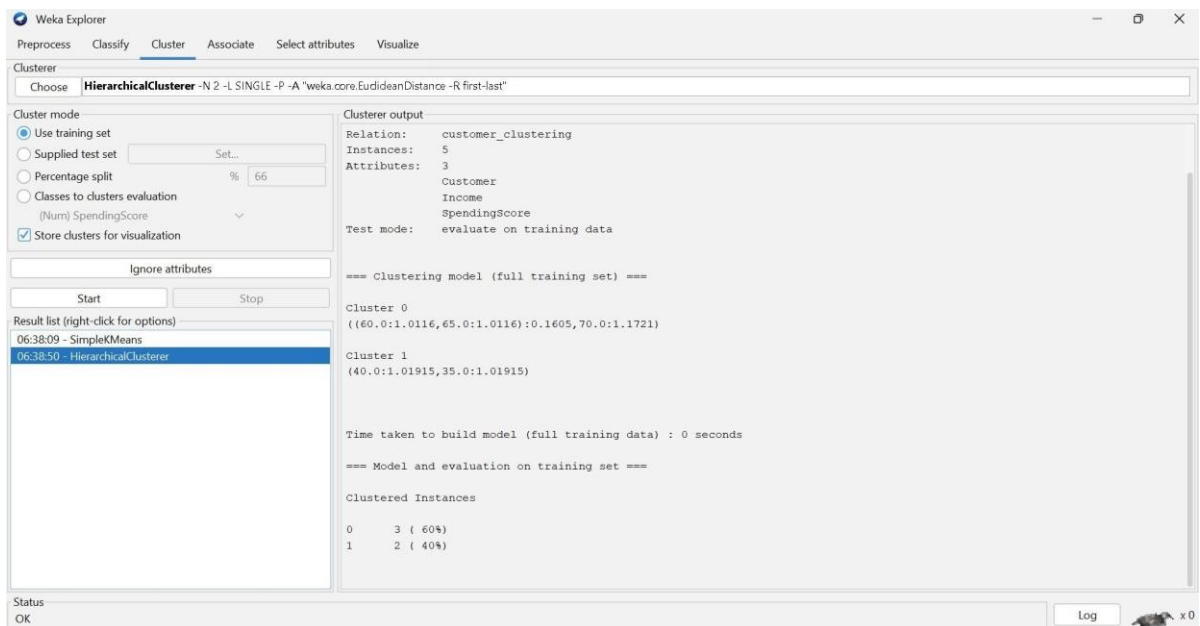
Files of Type: Arff data files (*.arff)

Open Cancel

Status

Welcome to the Weka Explorer

Log



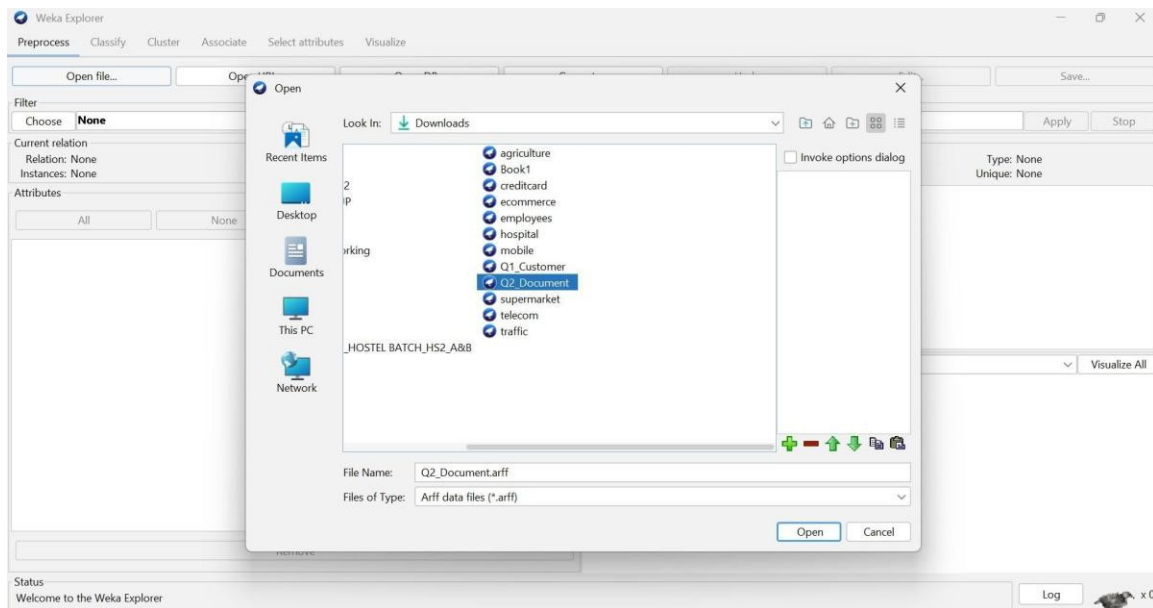
Q2. Dataset: Document Data

Document Word1 Word2 Word3

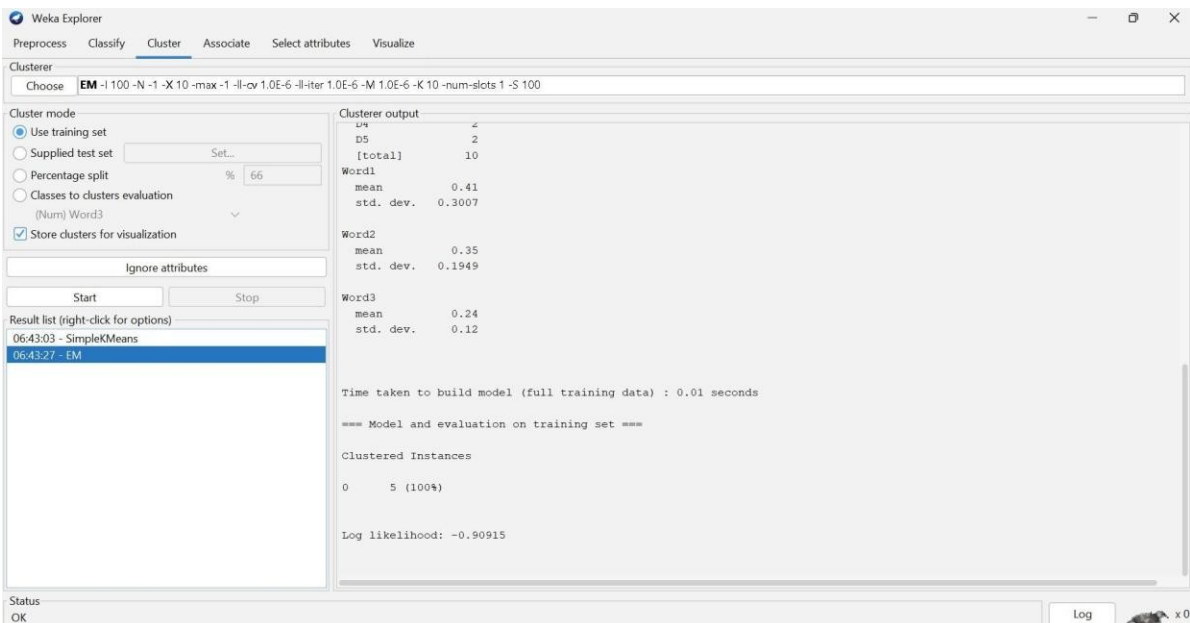
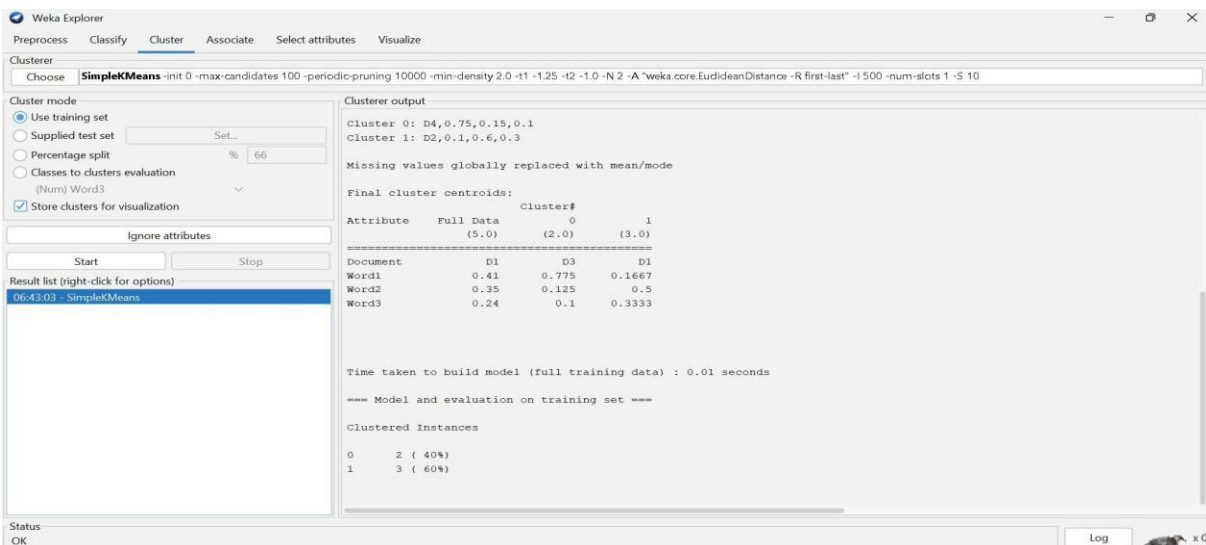
D1	0.2	0.5	0.3
D2	0.1	0.6	0.3
D3	0.8	0.1	0.1
D4	0.75	0.15	0.1
D5	0.2	0.4	0.4

Question:

Compare K-means and EM algorithms in handling overlapping clusters for the given dataset.
(BL4 – 1 Mark)



OUTPUT:



Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Clusterer: Choose **SimpleKMeans** -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 2 -A "weka.core.EuclideanDistance -R first-last" -I 500 -num-slots 1 -S 10

Cluster mode

- ☒ Use training set
- ☐ Supplied test set
- ☐ Percentage split
- ☐ Classes to clusters evaluation
- ☒ Store clusters for visualization

Ignore attributes

Start Stop

Result list (right-click for options)

06:43:03 - SimpleKMeans

Cluster output

Cluster 0: D4,0.75,0.15,0.1
Cluster 1: D2,0.1,0.6,0.3

Missing values globally replaced with mean/mode

Final cluster centroids:

Attribute	Full Data	Cluster#	0	1
	(5.0)	(2.0)	(3.0)	
Document	D1	D3	D1	
Word1	0.41	0.775	0.1667	
Word2	0.35	0.125	0.5	
Word3	0.24	0.1	0.3333	

Time taken to build model (full training data) : 0.01 seconds

=== Model and evaluation on training set ===

Clustered Instances

0	2 (40%)
1	3 (60%)

Status OK Log

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Clusterer: Choose **EM** -I 100 -N -1 -X 10 -max -1 -ll-cv 1.0E-6 -ll-iter 1.0E-6 -M 1.0E-6 -K 10 -num-slots 1 -S 100

Cluster mode

- ☒ Use training set
- ☐ Supplied test set
- ☐ Percentage split
- ☐ Classes to clusters evaluation
- ☒ Store clusters for visualization

Ignore attributes

Start Stop

Result list (right-click for options)

06:43:03 - SimpleKMeans
06:43:27 - EM

Cluster output

	4
D5	2
[total]	10
Word1	
mean	0.41
std. dev.	0.3007
Word2	
mean	0.35
std. dev.	0.1949
Word3	
mean	0.24
std. dev.	0.12

Time taken to build model (full training data) : 0.01 seconds

=== Model and evaluation on training set ===

Clustered Instances

0	5 (100%)
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Log likelihood: -0.90915

Status OK Log

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Clusterer: Choose **EM** -I 100 -N -1 -X 10 -max -1

Cluster mode

- ☒ Use training set
- ☐ Supplied test set
- ☐ Percentage split
- ☐ Classes to clusters evaluation
- ☒ Store clusters for visualization

Ignore attributes

Start Stop

Result list (right-click for options)

06:43:03 - SimpleKMeans
06:43:27 - EM

Weka Clusterer Visualize: 06:43:27 - EM (document_clustering)

X: Instance_number (Num) Y: Document (Nom)

Colour: Cluster (Nom) Select Instance

Reset Clear Open Save Jitter

Plot: document_clustering_clustered

Class colour cluster0

Status OK Log

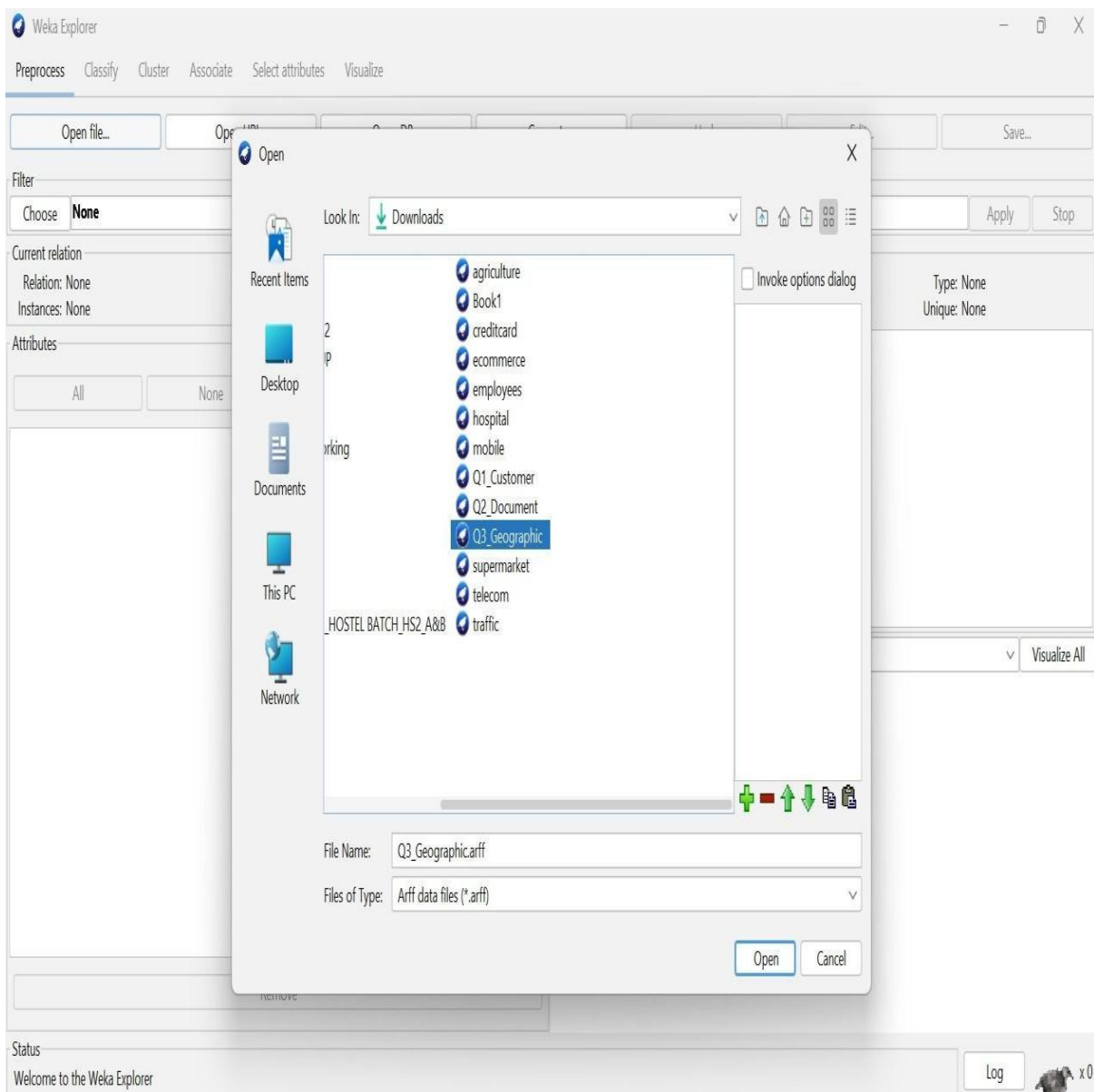
Q3. Dataset: Geographic Data

Point Latitude Longitude

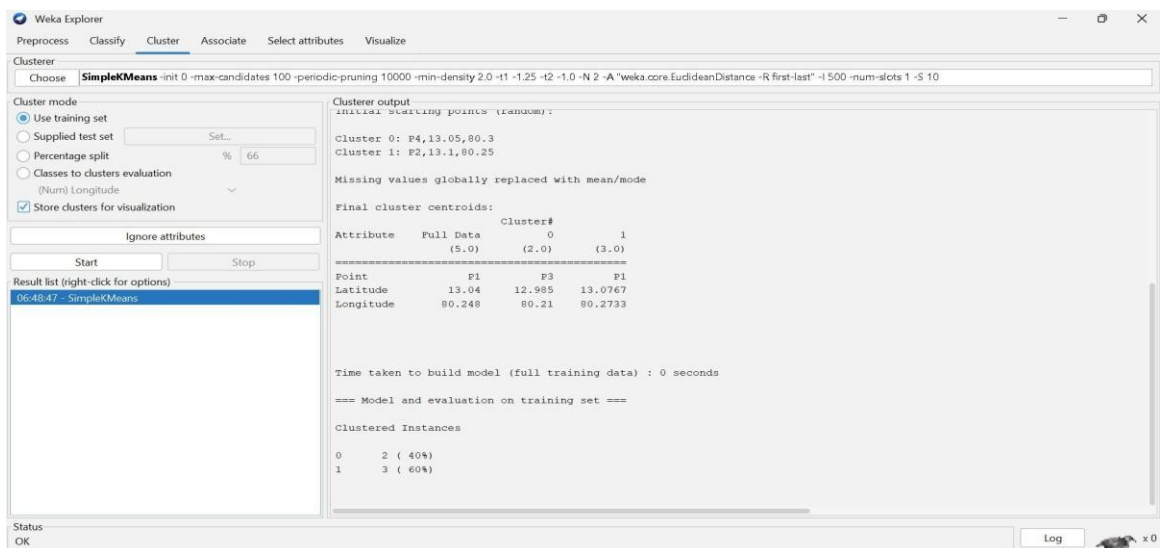
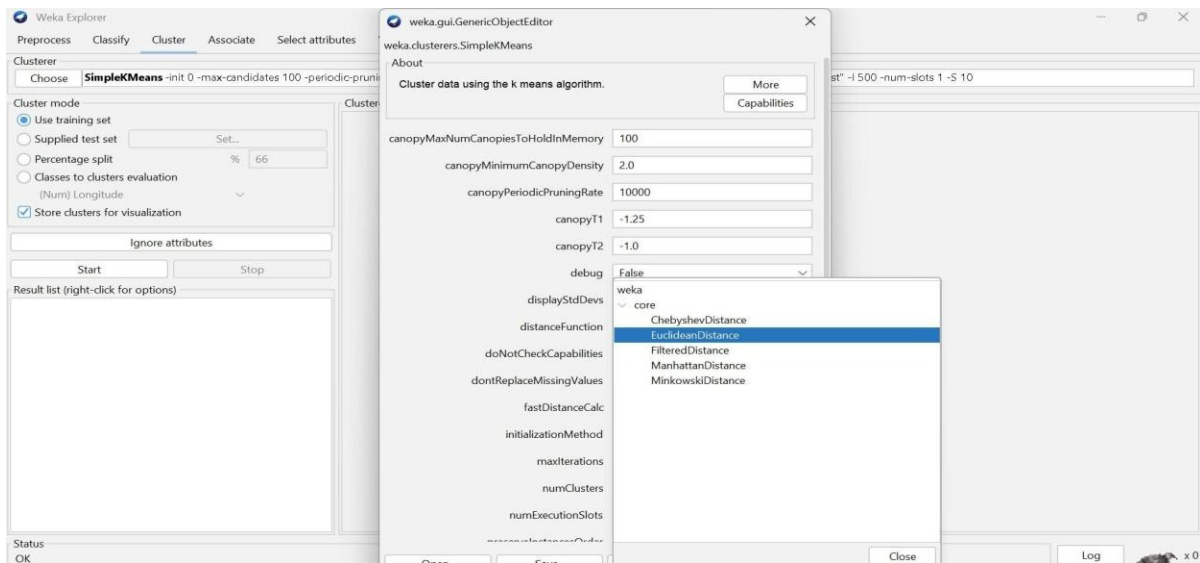
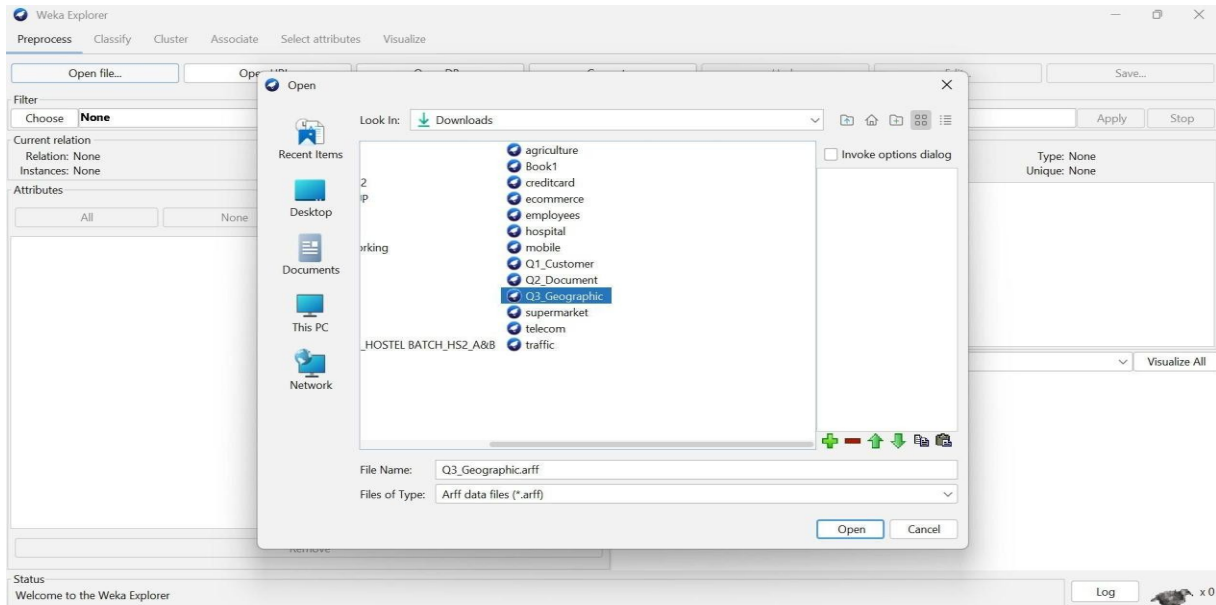
P1	13.08	80.27
P2	13.10	80.25
P3	12.97	80.22
P4	13.05	80.30
P5	13.00	80.20

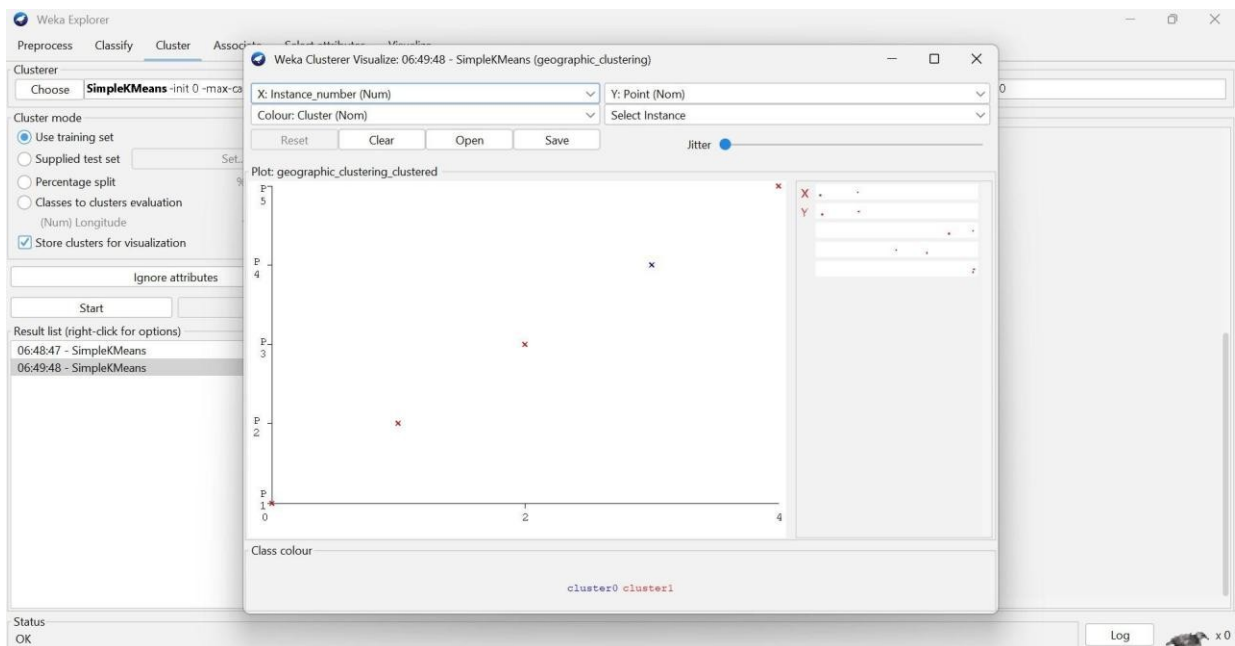
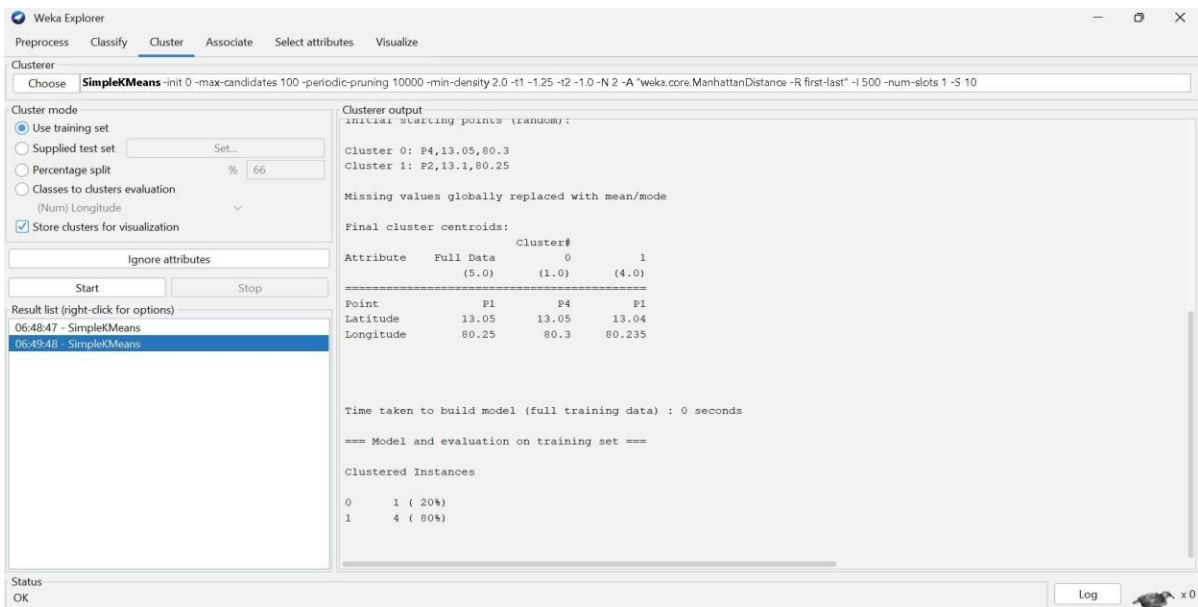
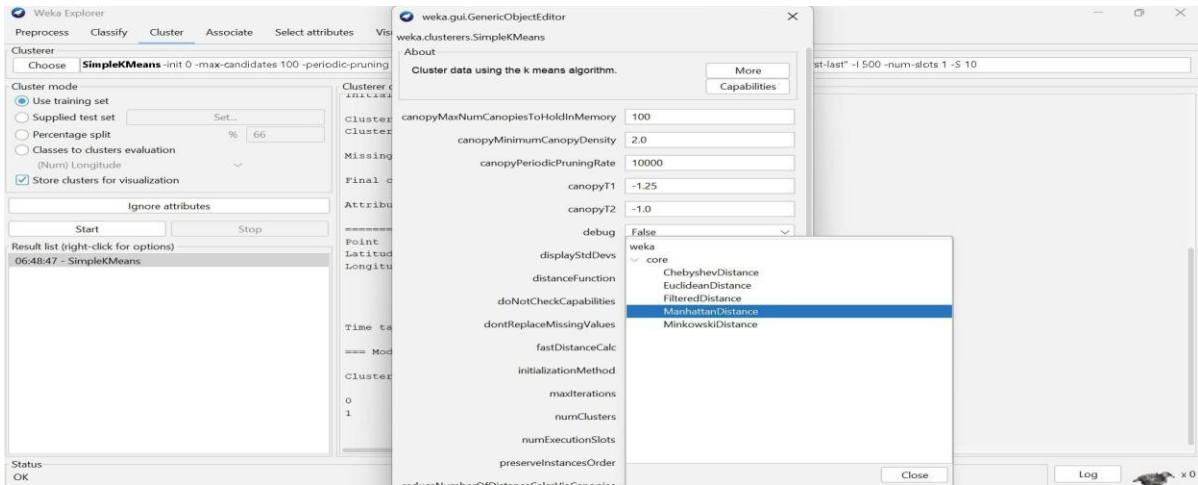
Question:

Compare Euclidean and Manhattan distance metrics and analyze their impact on clustering results. (BL4 – 1 Mark)



OUTPUT:





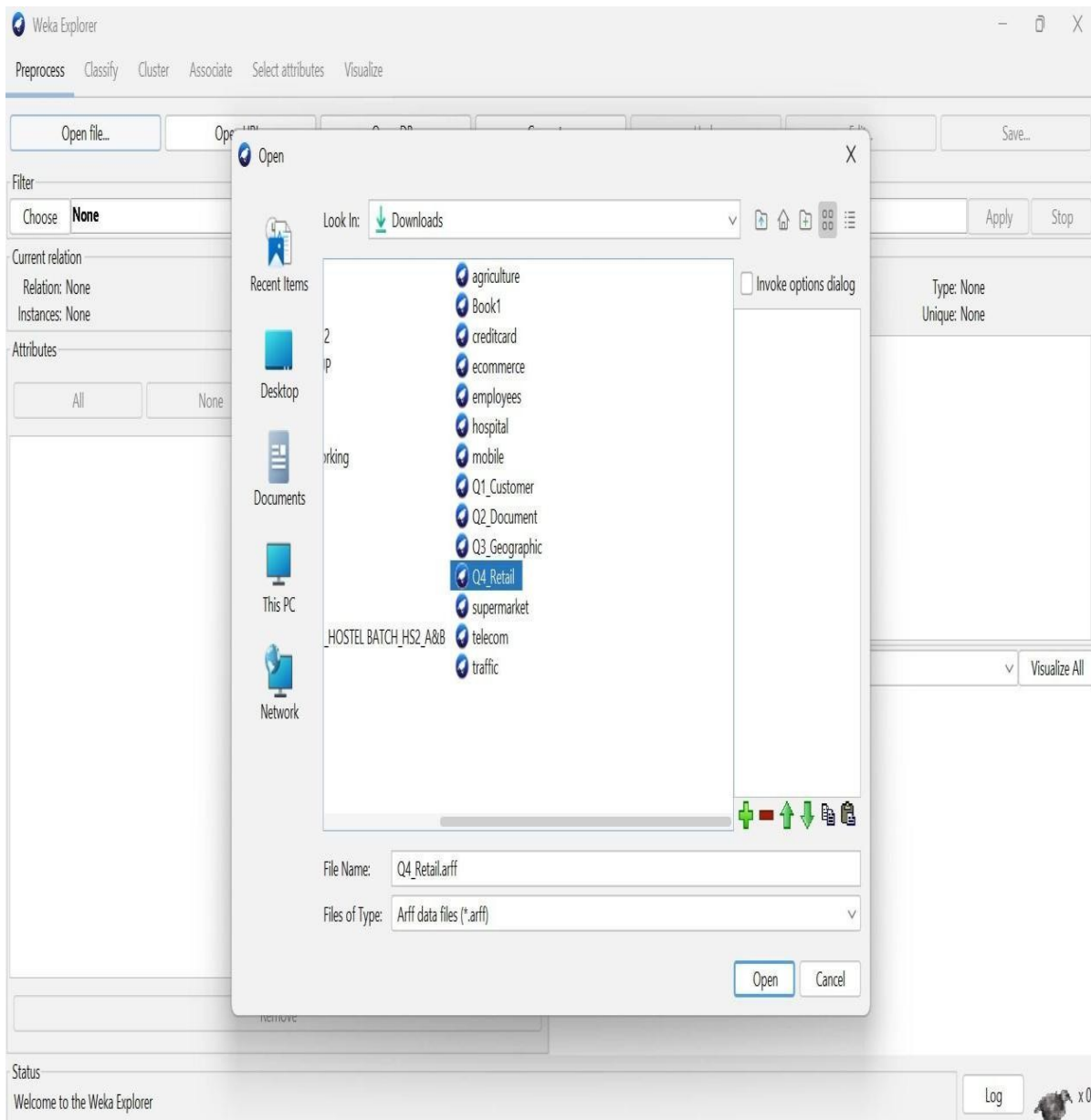
Q4. Dataset: Retail Products

Product Sales (₹) Frequency

P1	50000	20
P2	30000	10
P3	52000	22
P4	25000	8
P5	32000	12

Question:

Compare agglomerative and divisive clustering approaches and explain their suitability for this dataset. (BL3 – 1 Mark)



OUTPUT:

The screenshot shows the Weka Explorer application window. The 'Cluster' tab is selected. The 'Clusterer' dropdown is set to 'HierarchicalClusterer -N 2 -L SINGLE -P -A "weka.core.EuclideanDistance -R first-last"'. The 'Cluster mode' section has 'Use training set' selected. The 'Clusterer output' pane displays the following information:

Relation: retail_products
Instances: 5
Attributes: 3
Product
Sales
Frequency
Test mode: evaluate on training data

=== Clustering model (full training set) ===

Cluster 0
(20.0:1.01286,22.0:1.01286)

Cluster 1
((10.0:1.01286,12.0:1.01286):0.01412,8.0:1.02699)

Time taken to build model (full training data) : 0 seconds

=== Model and evaluation on training set ===

Clustered Instances

0	2 (40%)
1	3 (60%)

The 'Result list' on the left shows '06:53:39 - HierarchicalClusterer' as the selected result. The 'Status' bar at the bottom indicates 'OK'.

Weka Explorer

Preprocess **Cluster** Associate Select attributes Visualize

Clusterer: Choose **MakeDensityBasedClusterer** -M 1.0E-6 -W weka.clusterers.SimpleKMeans -- -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 2 -A "weka.core.EuclideanDistance -R first-last" -I 500 -n

Cluster mode:

- ☒ Use training set
- ☐ Supplied test set
- ☐ Percentage split %
- ☐ Classes to clusters evaluation (Num) Frequency
- ☒ Store clusters for visualization

Ignore attributes

Start

Result list (right-click for options):

- 06:53:39 - HierarchicalClusterer
- 06:54:50 - MakeDensityBasedClusterer**

Cluster output:

```

Discrete Estimator. Counts = 1 2 1 2 2 (Total = 8)
Attribute: Sales
Normal Distribution. Mean = 29000 StdDev = 2943.9203
Attribute: Frequency
Normal Distribution. Mean = 10 StdDev = 1.633

Cluster: 1 Prior probability: 0.4286

Attribute: Product
Discrete Estimator. Counts = 2 1 2 1 1 (Total = 7)
Attribute: Sales
Normal Distribution. Mean = 51000 StdDev = 1000
Attribute: Frequency
Normal Distribution. Mean = 21 StdDev = 1

Time taken to build model (full training data) : 0 seconds

=== Model and evaluation on training set ===

Clustered Instances

0      3 ( 60%)
1      2 ( 40%)

Log likelihood: -12.6953

```

Status: OK x 0

Weka Explorer

Preprocess **Cluster** Associate Select attributes Visualize

Clusterer: Choose **MakeDensityBasedClusterer** -M 1.0E-6 -W weka.clusterers.SimpleKMeans -- -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 2 -A "weka.core.EuclideanDistance -R first-last" -I 500 -n

Cluster mode:

- ☒ Use training set
- ☐ Supplied test set
- ☐ Percentage split %
- ☐ Classes to clusters evaluation (Num) Frequency
- ☒ Store clusters for visualization

Ignore attributes

Start

Result list (right-click for options):

- 06:53:39 - HierarchicalClusterer
- 06:54:50 - MakeDensityBasedClusterer**

Weka Clusterer Visualize: 06:54:50 - MakeDensityBasedClusterer (retail_products)

X: Instance_number (Num) Y: Product (Nom)

Colour: Cluster (Nom) Select Instance

 Jitter ☒

Plot: retail_products_clustered

Class colour: cluster0 cluster1

Status: OK x 0

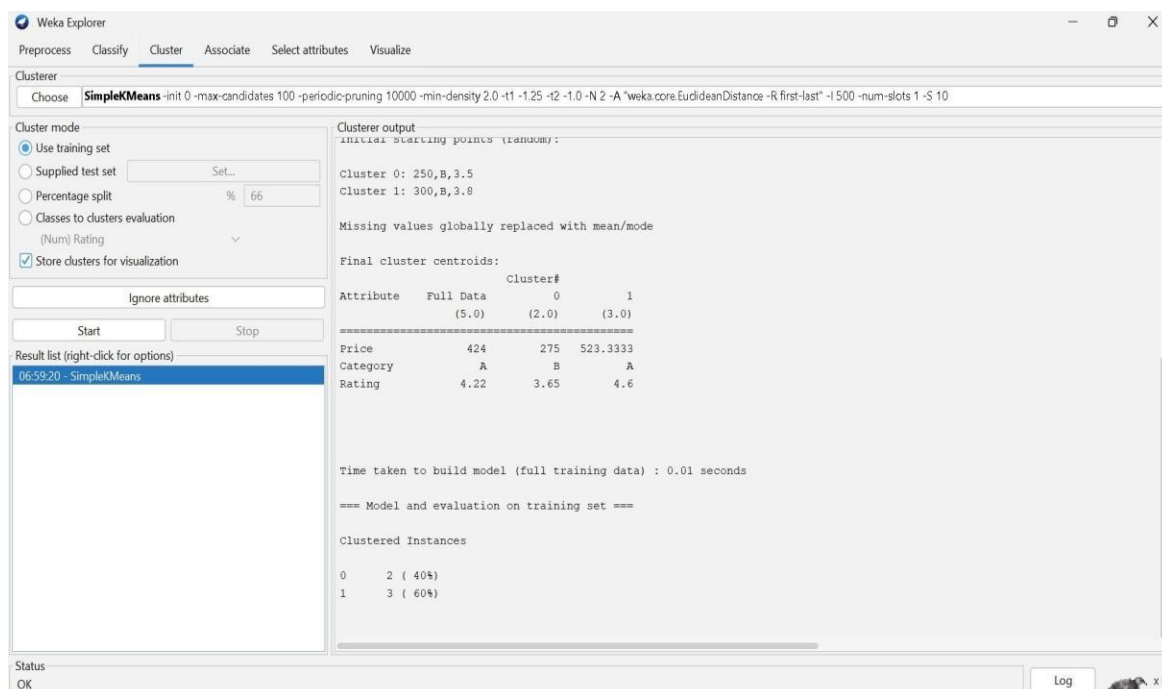
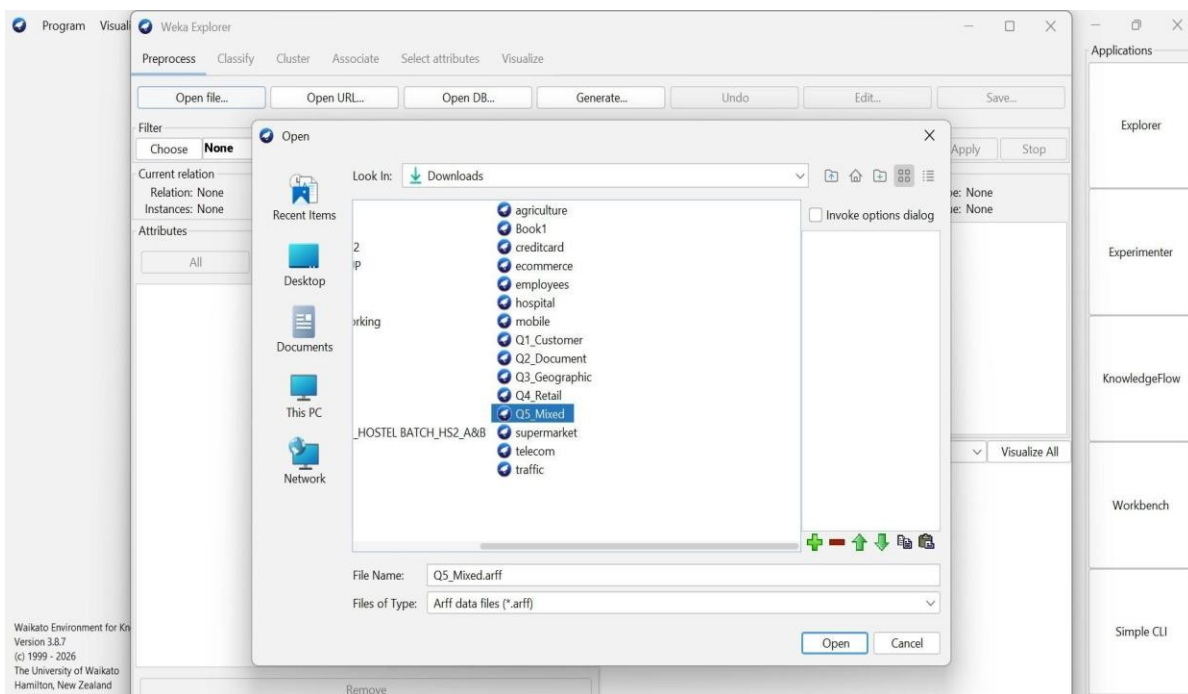
Q5 . Dataset: Mixed Attributes

Item Price (₹) Category Rating

I1	500	A	4.5
I2	300	B	3.8
I3	550	A	4.7
I4	250	B	3.5
I5	520	A	4.6

Question:

Compare centroid-based and density-based clustering techniques and analyze their effectiveness on mixed data types. (BL4 – 1 Mark)



OUTPUT:

